

Credit Card Approval Prediction

(by Using Machine Learning)

Project Description:

Credit Card Approval Prediction using Machine Learning is a web-based intelligent system developed to automate the credit card approval process. The application predicts whether a customer's credit card request is approved or rejected by analyzing applicant details such as income, employment status, age, loan amount, credit history, and other financial information. Different machine learning algorithms are trained and evaluated to identify the most accurate prediction model. The selected model is integrated into a Flask web application, allowing users to enter applicant information and receive instant approval predictions. This system reduces manual effort, improves consistency, and helps financial institutions make faster and more reliable decisions.

Problem Statement

Financial institutions receive thousands of credit card applications every day. Manual verification consumes significant time and may lead to inconsistent decisions due to human errors. An intelligent prediction system is required to automatically evaluate applicant information and accurately determine whether the application should be approved or rejected.

Scenarios:

Scenario 1

A customer applies for a credit card through the online portal. After entering personal and financial information, the system predicts whether the application is approved.

Scenario 2

A bank employee verifies an applicant with low annual income and poor credit history. The model predicts rejection based on previous training data.

Scenario 3

During peak application periods, the system automatically processes hundreds of applications and generates predictions within seconds, reducing manual workload.

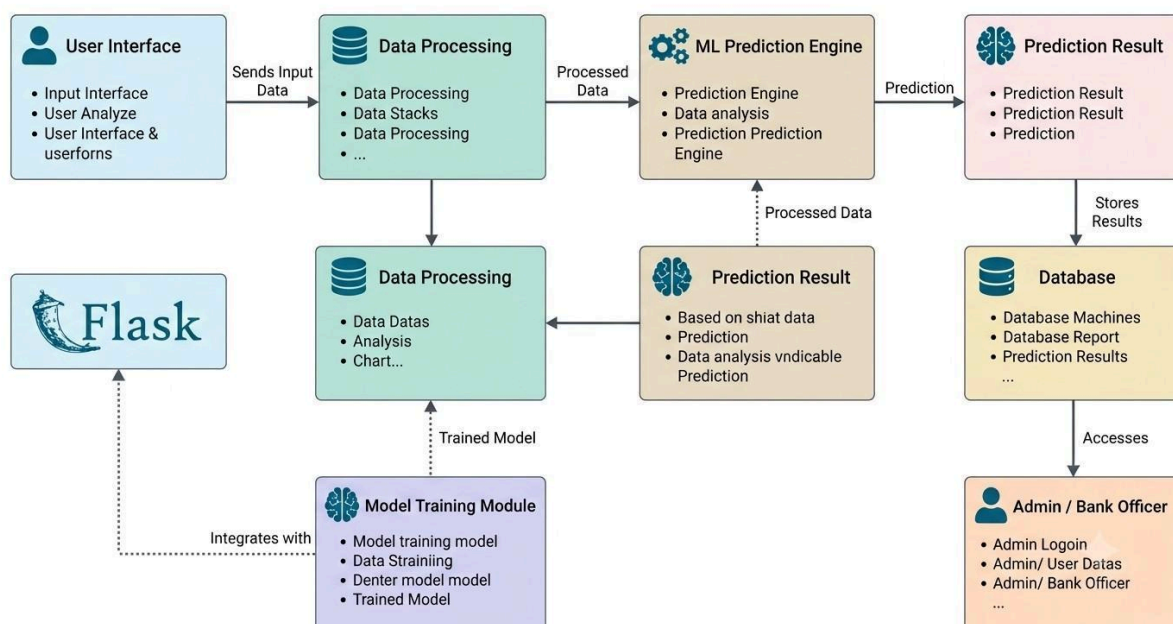
Objectives

- Develop a machine learning model for predicting credit card approval.
- Clean and preprocess applicant data.
- Compare multiple machine learning algorithms.

- ## Scope

Technical Architecture:

Architecture Flow

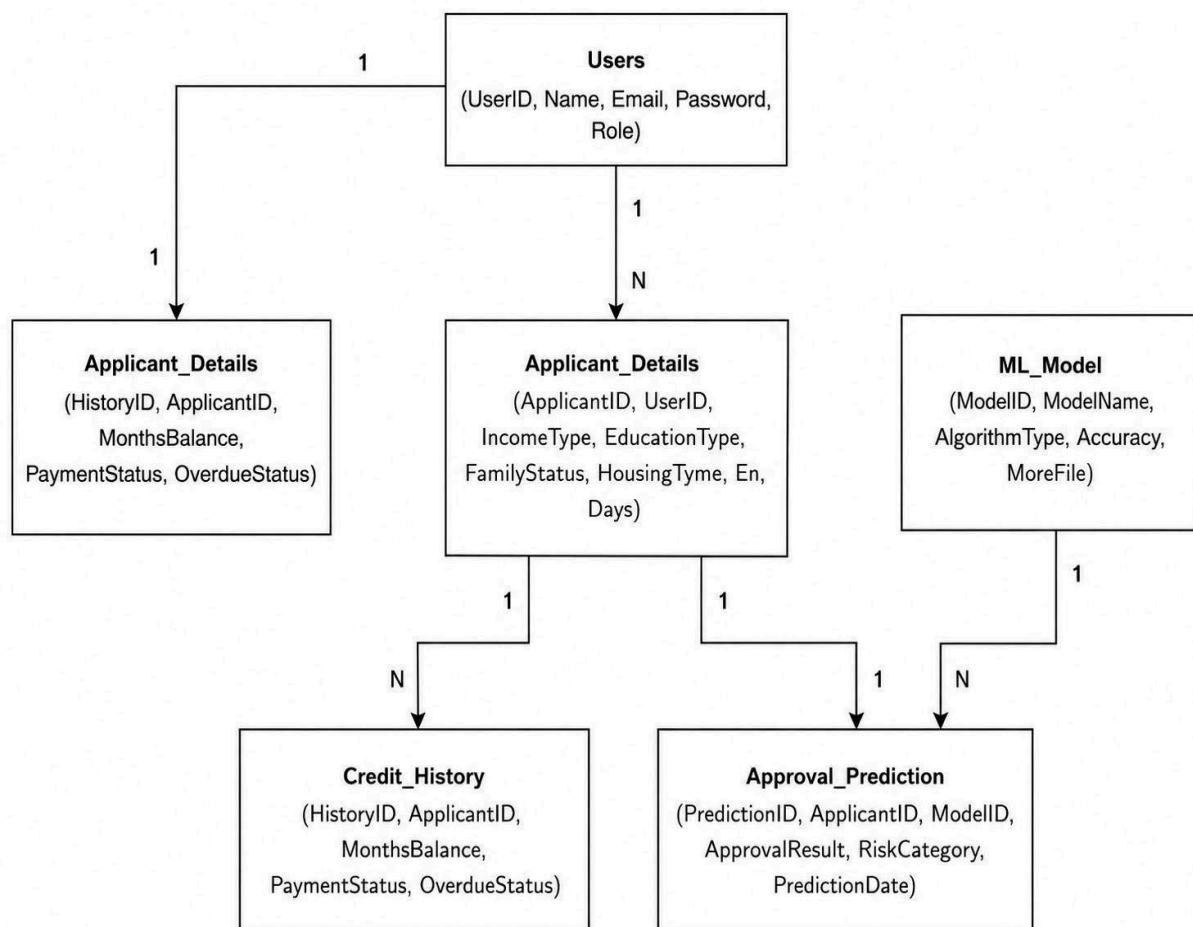


ER Diagram

Description:

The **Credit Card Approval Prediction System** follows a modular machine learning architecture to provide fast and accurate credit card approval decisions. Users enter applicant information

through a **Flask-based web application**, where the input data is validated, cleaned, encoded, and pre-processed before being sent to the **Machine Learning Prediction Engine**. The system evaluates the application using trained models such as **Logistic Regression, Decision Tree, Random Forest, and XGBoost**, with the best-performing model selected for real-time predictions. The prediction result (Approved or Rejected) is displayed instantly and stored for future reference and analysis. This architecture enables financial institutions to automate the credit card approval process, improve decision-making accuracy, reduce manual effort, and deliver a seamless user experience.



Pre-requisites

- Python Programming Proficiency (Python 3.x)
- Machine Learning Fundamentals — scikit-learn, XGBoost
- Data Handling Libraries — Pandas, NumPy
- Data Visualization — Matplotlib / Seaborn (for EDA)
- Web Framework Knowledge — Flask, for deployment of the prediction interface • Version Control with Git

Project Workflow:

Milestone 1

- Collect dataset
- Understand dataset
- Perform EDA

Milestone 2

- Handle missing values
- Encode categorical features
- Normalize data
- Split dataset

Milestone 3

- Train ML models
- Compare accuracy

Milestone 4

- Evaluate models
- Select best model

Milestone 5

- Save model
- Integrate Flask
- Deploy on Vercel

The application has been successfully deployed and is live at the following URL:

<https://sample-project-for-deploy-6sfb-lake.vercel.app/>

Full implementation details, source code, and screenshots of the deployed web application are provided in Section 6 (Web Application).

Methodology

Dataset Description

Attribute	Details
Dataset Name	clean_dataset.csv
Type	Structed
Target Variable	Credit Card Approval Status(accepted/rejected)
Feature Types	Categorical & Numerical attributes of attributes

Data Preprocessing Steps

- **Data Loading and Inspection:** The dataset (690 records, 16 columns) was loaded using Pandas, followed by inspection using `df.head()`, `df.tail()`, `df.info()`, and `df.describe()` to understand structure, data types, and summary statistics.
- **Missing Value Verification:** The dataset was checked for missing/null values prior to model training to ensure data completeness.
- **Duplicate Checking:** Duplicate records were identified and handled to prevent data leakage and bias.
- **Categorical Encoding:** Categorical features (Industry, Ethnicity, Citizen) were encoded using Label Encoding / One-Hot Encoding to convert them into a numerical format suitable for machine learning models.
- **Log Transformation:** The Income feature was log-transformed to reduce rightskewness caused by extreme high-income values (up to ₹100,000).
- **Feature Engineering:** A Debt-to-Income Ratio feature was derived to capture applicant financial risk more effectively.
- **Train-Test Split:** The dataset was divided into training (70%) and testing (30%) subsets using `train_test_split` from scikit-learn to evaluate model generalization.
- **Feature Scaling:** `StandardScaler` was applied to numerical features where required by the algorithm (e.g., Logistic Regression).

Algorithms Used

- Logistic Regression
- Decision Tree Captures

Evaluation Metrics

- **Accuracy** — Overall proportion of correctly classified applications.
- **Precision** — Proportion of predicted approvals that were actually correct.
- **Recall** — Proportion of actual approvals that were correctly identified.
- **F1-Score** — Harmonic mean of Precision and Recall, useful for imbalanced classes.
- **Confusion Matrix** — Detailed breakdown of true/false positives and negatives
- **ROC-AUC** — Measures the model's ability to distinguish between approved and rejected classes across thresholds

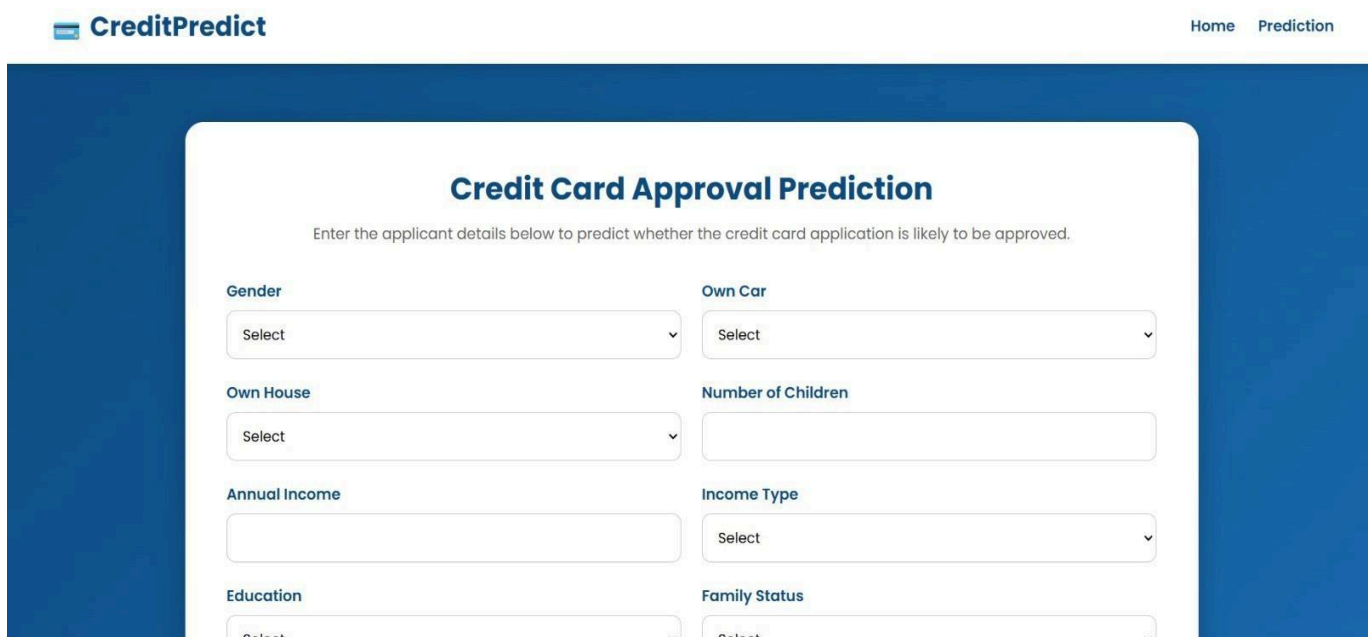
Web Application

The trained XGBoost model has been deployed as an interactive web application named "Credit Card Approval Prediction", built using Flask (backend) and HTML/CSS/JavaScript (frontend). The application allows a user to enter applicant details through a form and receive an instant Approved / Not Approved prediction along with a confidence percentage.

The application has been successfully deployed and is publicly accessible at:

<https://sample-project-for-deploy-6sfb-lake.vercel.app/>

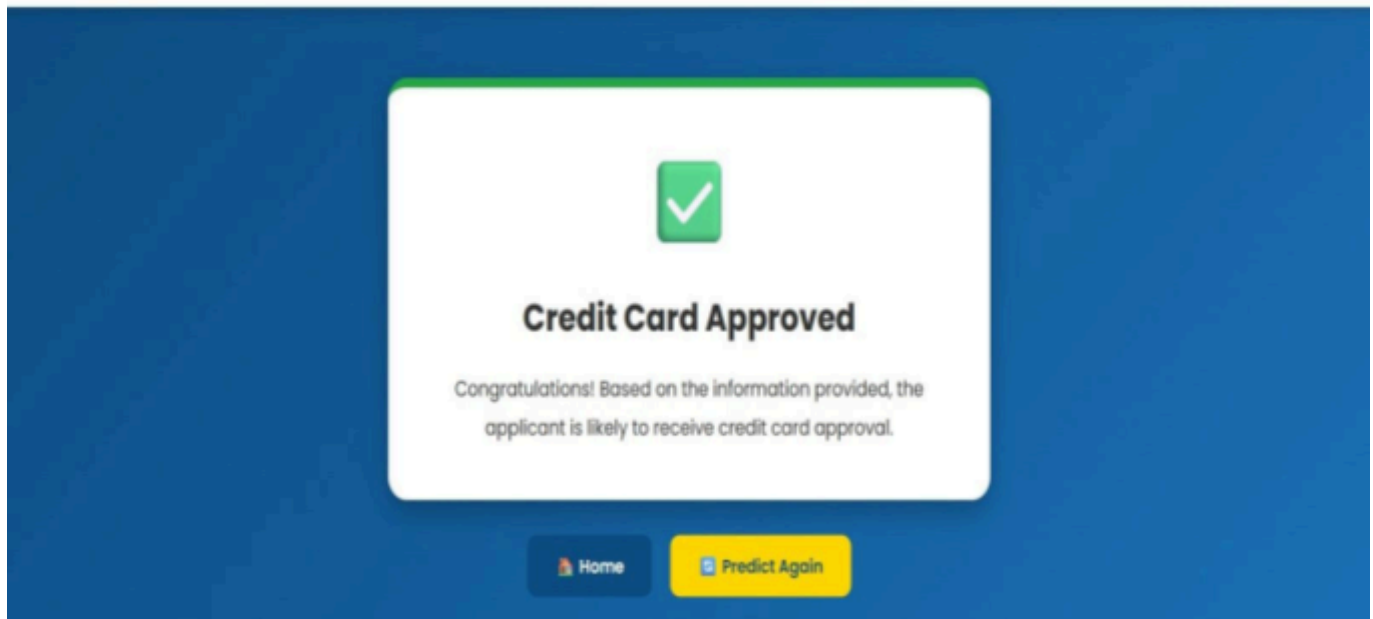
Application Form



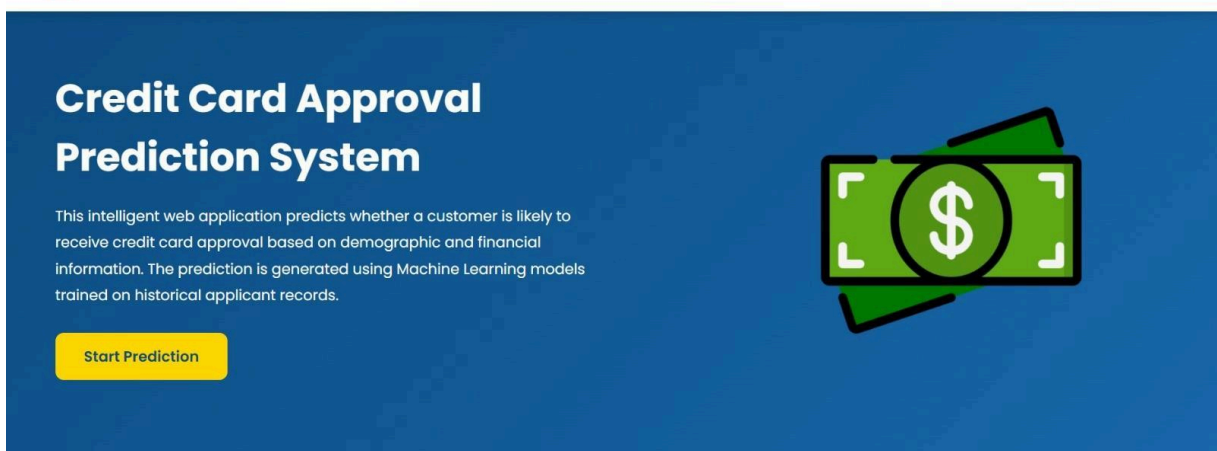
The screenshot shows a web application titled "CreditPredict" with a navigation bar containing "Home" and "Prediction" links. The main content area is titled "Credit Card Approval Prediction" and includes a subtitle: "Enter the applicant details below to predict whether the credit card application is likely to be approved." The form consists of several input fields arranged in two columns:

- Gender:** A dropdown menu with "Select" as the placeholder.
- Own Car:** A dropdown menu with "Select" as the placeholder.
- Own House:** A dropdown menu with "Select" as the placeholder.
- Number of Children:** A text input field.
- Annual Income:** A text input field.
- Income Type:** A dropdown menu with "Select" as the placeholder.
- Education:** A dropdown menu with "Select" as the placeholder.
- Family Status:** A dropdown menu with "Select" as the placeholder.

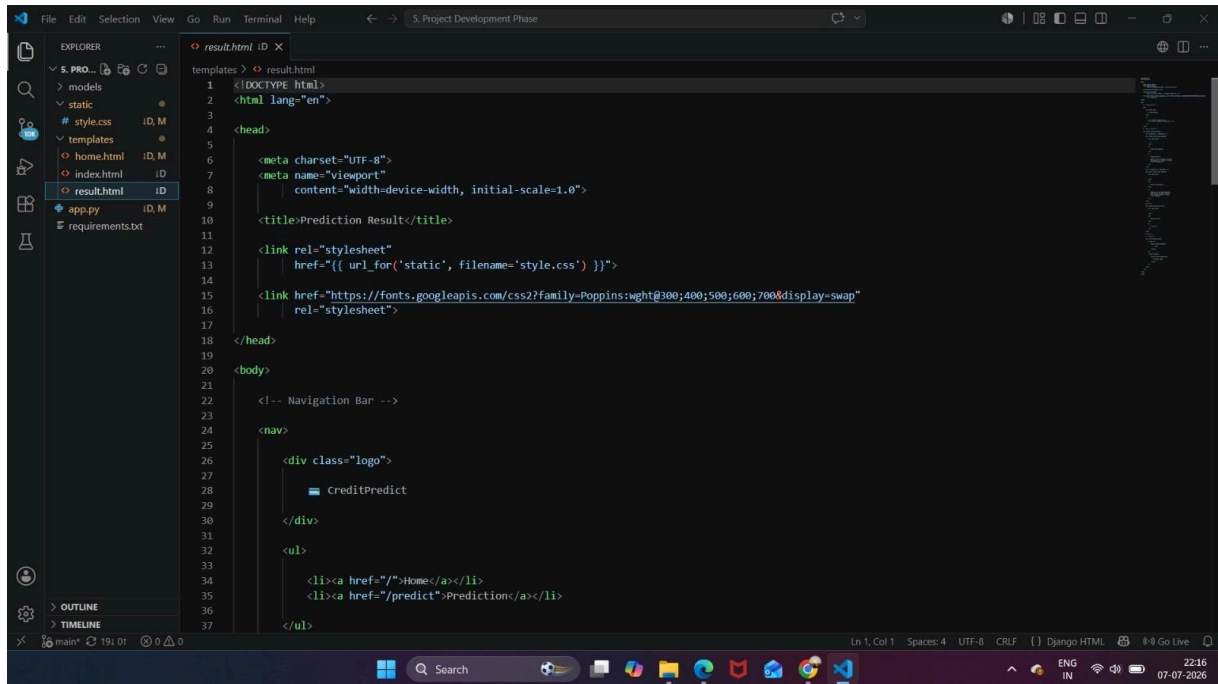
Prediction of Result

 CreditPredict[Home](#) [Prediction](#)

Interface

 CreditPredict[Home](#) [Prediction](#) CreditPredict[Home](#) [Prediction](#)

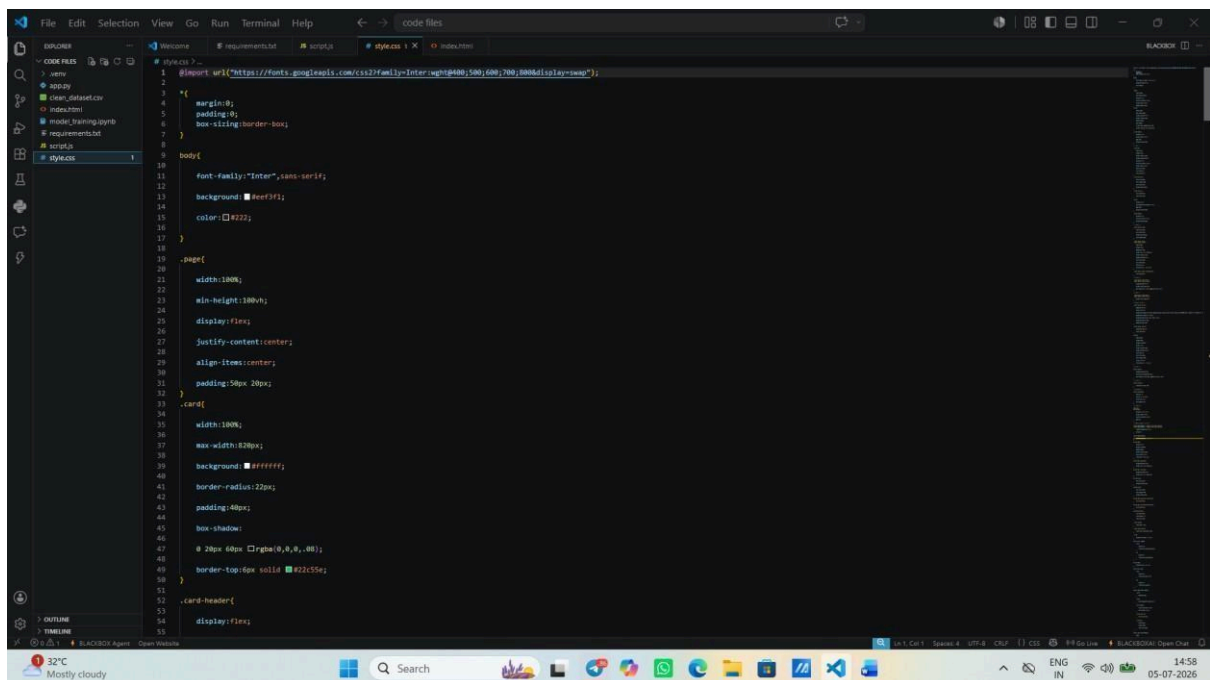
HTML(structure)



```

1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5
6   <meta charset="UTF-8">
7   <meta name="viewport"
8     content="width=device-width, initial-scale=1.0">
9
10  <title>Prediction Result</title>
11
12  <link rel="stylesheet"
13    href="{{ url_for('static', filename='style.css') }}">
14
15  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap"
16    rel="stylesheet">
17
18 </head>
19
20 <body>
21
22   <!-- Navigation Bar -->
23
24   <nav>
25
26     <div class="logo">
27       <img alt="CreditPredict logo" data-bbox="315 348 378 358"/>
28     </div>
29
30     <ul>
31       <li><a href="/">Home</a></li>
32       <li><a href="/predict">Prediction</a></li>
33     </ul>
34
35   </nav>
36
37
  
```

CSS



```

1 @import url("https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap");
2
3 {
4   margin: 0;
5   padding: 0;
6   box-sizing: border-box;
7 }
8
9 body {
10
11   font-family: "Inter", sans-serif;
12   background-color: #f0f0f0;
13   color: #333;
14 }
15
16 .page {
17
18   width: 100%;
19   min-height: 100vh;
20   display: flex;
21   justify-content: center;
22   align-items: center;
23   padding: 50px 20px;
24 }
25
26 .card {
27
28   width: 100%;
29   max-width: 820px;
30   background-color: #ffffff;
31   border-radius: 22px;
32   padding: 40px;
33   box-shadow: 0 20px 60px solid #d2d2d2;
34 }
35
36 .card-header {
37
38   display: flex;
39 }
  
```


Backend Implementation Flask(app.py)

```

File Edit Selection View Go Run Terminal Help 5. Project Development Phase
EXPLORER
  5. PROJECT DEVELOPMENT...
    > models
    > static
    > templates
    > app.py ID, M
    requirements.txt
  app.py ID, M X
    30 def result():
    31     work_phone,
    32     phone,
    33     email,
    34     family_members
    35     ))
    36
    37     # 4. Predict
    38     prediction = model.predict(features)[0]
    39
    40     # Fixed Variable conflict (changed variable 'result' to 'prediction_text')
    41     if prediction == 1:
    42         prediction_text = "Approved"
    43         color = "green"
    44     else:
    45         prediction_text = "Rejected"
    46         color = "red"
    47
    48     return render_template(
    49         "result.html",
    50         prediction=prediction_text,
    51         color=color
    52     )
    53
    54     except Exception as e:
    55         # Prints the error directly to your VS Code terminal for easier debugging
    56         print(f"Error submission error: {e}")
    57         return render_template(
    58             "result.html",
    59             prediction="Error",
    60             color="red",
    61             error=str(e)
    62         )
    63
    64     if __name__ == "__main__":
    65         app.run(debug=True)
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Conclusion and Future Scope

Conclusion

The Credit Card Approval Prediction using Machine Learning project successfully demonstrates how machine learning can be applied to automate and improve the credit card approval process. By analysing applicant information such as income, employment status, education, family status, housing type, and financial history, the developed model can accurately predict whether a credit card application is likely to be approved or rejected. Various classification algorithms were evaluated, and the best-performing model was integrated into a Flask-based web application to provide real-time predictions through a user-friendly interface. This system helps reduce manual effort, minimizes human bias, speeds up decision-making, and supports financial institutions in making consistent and reliable credit approval decisions. Overall, the project proves that machine learning is an effective solution for enhancing the efficiency, accuracy, and transparency of the credit card approval process.

Future Scope

The proposed Credit Card Approval Prediction system can be further enhanced with several advanced features to improve its performance and real-world usability. Future improvements include:

- Integrating larger and more diverse datasets to improve model accuracy and generalization.
- Deploying the application on cloud platforms such as AWS, Azure, or Google Cloud for better scalability and accessibility.
- Implementing deep learning techniques and ensemble learning methods to further enhance prediction performance.
- Integrating real-time credit score and banking APIs to make predictions using live financial data.
- Incorporating fraud detection mechanisms to identify suspicious or fraudulent credit card applications.
- Developing a mobile application for easier access by customers and bank officials.
- Implementing role-based authentication and enhanced security measures to protect sensitive customer information.
- Extending the system to support loan approval, insurance eligibility, and other financial risk assessment applications using the same machine learning framework.